

Article

Effect of Sintering Temperature on the Microstructure and Integrated Properties of MgAlTiVFeCo Lightweight High-Entropy Alloy

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Abstract

To develop and design new alloy materials with lightweight and superior comprehensive performance. In this study, a MgAlTiVFeCo lightweight high-entropy alloy (LW-HEA) was fabricated via mechanical alloying and spark plasma sintering (SPS) to investigate the effects of sintering temperature on its phase structure, microstructure, densification, microhardness, high-temperature oxidation resistance, and corrosion resistance. The results indicate that the ball-milled MgAlTiVFeCo LW-HEA formed a simple solid solution phase with a BCC structure. After spark plasma sintering, the phase structure of the alloy changed, maintaining the BCC phase as the primary phase while accompanying the precipitation of secondary phases. When the sintering temperature reached 1000 °C, the alloy achieved a densification of 96.7% and a microhardness of 1235.5 HV. Its hardness value is comparable to the typical range of cemented carbides, demonstrating outstanding mechanical properties. The oxidation kinetics of MgAlTiVFeCo high-entropy alloys sintered at different temperatures at 900 °C follow a parabolic law, which is diffusion-controlled and can be divided into two stages: rapid growth and slow stabilization. At a sintering temperature of 1000 °C, the fitted oxidation rate constants, k_{p1} (0–25 h) and k_{p2} (25–60 h), are $3.76 \times 10^{-2} \text{ mg}^2 \cdot \text{cm}^{-4} \cdot \text{s}^{-1}$ and $1.10 \times 10^{-1} \text{ mg}^2 \cdot \text{cm}^{-4} \cdot \text{s}^{-1}$, respectively, outperforming those of alloys sintered at other temperatures. In a 3.5 wt% NaCl solution, the corrosion resistance of the alloy improves with increasing sintering temperature. Compared to alloys sintered at medium-to-low temperatures (850–950 °C), the alloy sintered at a high temperature (1000 °C) exhibits a more positive corrosion potential (−0.438 V) and a lower corrosion current density ($1.07 \times 10^{-6} \text{ A} \cdot \text{cm}^{-2}$), indicating excellent corrosion resistance. It is evident that 1000 °C is the optimal sintering temperature, and the MgAlTiVFeCo LW-HEA demonstrates superior comprehensive properties.

Keywords: high-entropy alloy; microstructure; mechanical properties; high temperature oxidation resistance; corrosion resistance



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1. Introduction

With the advancement of society and continuous progress in science and technology, the room for improving the performance of traditional alloys has become increasingly

limited, making it difficult to meet the ever more stringent demands for comprehensive material properties in cutting-edge equipment. To break through this bottleneck, high-entropy alloys [1] have been proposed as a revolutionary materials design concept. Composed of four or more principal elements in equiatomic or near-equiatomic ratios, their high mixing entropy effect suppresses the formation of brittle intermetallic compounds and promotes the formation of simple solid-solution phases. This unique multi-principal-element, high-entropy composition endows high-entropy alloys with a series of exceptional properties that are difficult to achieve simultaneously in conventional alloys, such as high strength, high hardness, excellent wear resistance, good thermal stability, high-temperature oxidation resistance, and corrosion resistance [2–6]. Consequently, they have rapidly become a research frontier in high-performance structural materials. Driven by the urgent need for weight reduction and efficiency improvement in fields such as aerospace and rail transportation, the development of lightweight high-entropy alloys (LW-HEAs) that combine low density with outstanding mechanical properties, corrosion resistance, and high-temperature oxidation resistance has emerged as an important and highly promising direction in this field.

Currently, research on lightweight high-entropy alloys primarily focuses on alloy systems based on low-density elements such as Mg, Al, Li, Be, Ti, Si, and Ca as principal components. The aim is to utilize the design concept of high-entropy alloys to reduce density while retaining their excellent properties, thereby developing novel high-performance materials with densities below 6 g/cm³ [7]. As the lightest metallic structural material (density 1.74 g/cm³), the introduction of Mg plays an indispensable role in significantly reducing alloy density. In reported literature, the actual density of Mg-containing high-entropy alloys ranges from 1.46 g/cm³ to 6 g/cm³ [8]. Among these, Al₁₅Li₃₉Mg₄₅Ca_{0.5}Si_{0.5} [9] and Al₁₀Mg₃₀Ti₂₅Zr₁₀Hf₂₅ [10] represent the alloys with the lowest and highest densities, respectively, within Mg-containing HEAs. With in-depth research on lightweight high-entropy alloys, it has been found that when designing LW-HEAs using lightweight elements, especially Mg as the principal component, the alloys often exhibit complex microstructures containing various intermetallic compounds, predominantly manifesting as solid solution + intermetallic compound (SS + IM)-type lightweight high-entropy alloys [11]. Nevertheless, they retain the excellent properties of high-entropy alloys, such as high hardness, high strength, high oxidation resistance, and high corrosion resistance. Shao et al. [12] prepared six kinds of lightweight high-entropy alloys rich in Al and Mg elements using induction melting, with densities ranging between 2.64 and 2.75 g/cm³, all exhibiting SS + IM microstructures. The Al₂₀Be₂₀Fe₁₀Si₁₅Ti₃₅ lightweight high-entropy alloy designed by Tseng et al. [13] has a density of only 3.91 g/cm³ but a hardness as high as 911 HV, along with excellent oxidation resistance at 700 °C and 900 °C. Yao et al. [14] employed high-throughput gradient continuous casting technology to prepare a compositionally graded MgAlCuZnGd lightweight high-entropy alloy and systematically investigated its mechanical properties, microstructure, and corrosion behavior across different gradient regions. The results indicate that the alloy mainly consists of α -Mg, Al₂Gd, MgCuZn, Mg₂Cu, and (Mg,Zn)₃Gd phases. Region V (Mg_{79.46}Al_{4.49}Cu_{5.92}Zn_{5.80}Gd_{4.94}) achieved a compressive strength of 474 MPa with approximately 14.0% plasticity. Region VI (Mg_{79.18}Al_{4.48}Cu_{5.89}Zn_{6.88}Gd_{3.58}) exhibited the best corrosion resistance. Satya et al. [15] successfully fabricated AlMg_xTiSnZn ($x = 0, 0.5, 1.0, 1.5$) lightweight high-entropy alloys via mechanical alloying and spark plasma sintering, investigating the relationship between their microstructure, phase stability, and functional properties (mechanical, thermal, electrochemical, and oxidation resistance). The results show that the alloy densities range from 4.31 to 5.63 g/cm³, with the BCC phase as the primary constituent alongside an appropriate

amount of intermetallic compounds. When $x = 1.0$, the alloy demonstrates high microhardness, high fracture strength, favorable specific strength, and excellent corrosion resistance.

While research on magnesium-containing lightweight high-entropy alloys (Mg-containing LWHEAs) has made initial progress, significant limitations remain in understanding their microstructural evolution and functional properties. On one hand, the subsequent phase transformation pathways, precipitation sequences, and kinetic mechanisms of metastable solid solution phases under non-equilibrium processing conditions are not yet clear. On the other hand, systematic studies are lacking on the segregation behavior, interfacial energy, and diffusion characteristics of light elements and transition metals at interfaces, with a particular shortage of integrated atomic-scale experiments and simulations. Furthermore, existing work has primarily focused on fundamental mechanical properties such as hardness and room-temperature compression, while research on microstructural evolution laws and key functional properties like high-temperature oxidation resistance and corrosion resistance remains relatively scarce. Additionally, the significant differences in atomic radius, electronegativity, and other properties between magnesium and other constituent elements often lead to the formation of stable intermetallic compounds, further constraining the optimization of material performance. To address these challenges, this study employs mechanical alloying (MA)—a method capable of extending solid solubility and suppressing intermetallic compound formation [16]—combined with spark plasma sintering (SPS) to fabricate a novel six-component MgAlTiVFeCo lightweight high-entropy alloy. The effects of sintering temperature on the phase composition, microstructure, densification behavior, microhardness, high-temperature oxidation resistance, and corrosion resistance of the alloy are systematically investigated. This work aims to provide experimental evidence and theoretical reference for developing high-performance lightweight high-entropy alloys through composition and process design.

2. Materials and Methods

The raw materials used in the experiment include elemental powders of Al (purity $\geq 99.0\%$, particle size 150–250 mesh), Ti (purity $\geq 99.0\%$, particle size 200–300 mesh), V (purity $\geq 99.5\%$, particle size ≥ 300 mesh), Fe (purity 99.0% , particle size 300 mesh), and Co (purity $\geq 99.0\%$, particle size 200 mesh), as well as MgAl alloy powder (purity 99.0% , particle size 120–230 mesh, Mg-to-Al mass ratio 1:1). Among these, the elemental powders of Al, Ti, Fe, and Co were supplied by Sinopharm Chemical Reagent Co., Ltd. (Shanghai, China), the V powder was provided by Jinzhou Haixin Metal Materials Co., Ltd. (Jinzhou, China), and the MgAl alloy powder was obtained from Shanxi Fusheng Magnesium Industry Co., Ltd. (Yuncheng, China). These powders were weighed and mixed in equimolar proportions of Mg, Al, Ti, V, Fe, and Co. Wet milling was performed using an XQM-4 vertical planetary ball mill (Changsha Miqi Instrument Equipment Co., Ltd., Changsha, China) with a stainless steel milling jar and stainless steel grinding balls of different diameters (15 mm, 10 mm, 5 mm, mass ratio 2:3:5). The ball-to-powder ratio was set at 10:1, the rotational speed was 400 r/min, and anhydrous ethanol was used as the process control agent. Given the extremely high chemical reactivity of magnesium, multiple strategies were employed synergistically throughout the milling process to control its oxidation tendency; for example, MgAl pre-alloyed powder was used instead of pure magnesium powder to reduce the surface reactivity of magnesium via pre-alloying; after loading the powders, anhydrous ethanol was added to the milling jar for wet milling. The covering and cooling effects of ethanol further slowed the reaction between fresh powder surfaces and residual oxygen; a high ball-to-powder ratio (10:1) was adopted to enhance alloying efficiency, shorten the total time required to achieve sufficient alloying, and thereby indirectly reduce the exposure time of the powder in its active state to a potentially oxidiz-

ing environment [17,18]. Nevertheless, completely avoiding oxygen introduction during prolonged high-energy mechanical alloying remains highly challenging. The repeated fracturing and cold welding processes experienced by powder particles continuously generate large amounts of fresh atomic surfaces. Trace residual oxygen in the jar, adsorbed moisture and gases on the powder itself, and even minor reactive components that may decompose from the process control agent can all serve as sources of oxygen. The milling time was set to 100 h, and samples were taken after 20, 40, 80, and 100 h of milling to investigate the evolution of the powders during the process.

The ball-milled powder was sieved through a 300-mesh standard sieve to remove any possible coarse agglomerates, thereby obtaining alloy powder for subsequent sintering. MgAlTiVFeCo lightweight high-entropy alloys (LW-HEAs) were prepared by spark plasma sintering (SPS). The sintering process was carried out under a vacuum maintained at 10^{-2} Pa. The temperature was first raised to 600 °C at a heating rate of 150 °C/min and held for 5 min, then increased to the sintering temperature at a rate of 100 °C/min. Under a constant axial pressure of 25 MPa, the samples were held at the sintering temperature for 10 min, followed by furnace cooling to room temperature. The theoretical melting point of MgAlTiVFeCo was calculated to be 1320 °C, and the sintering temperature is generally selected within 0.6–0.9 times the theoretical melting point [19]. For the nanocrystalline powder obtained by high-energy ball milling in this system, when the SPS temperature was lower than 800 °C, the sintering driving force was insufficient, resulting in compacts with very low relative density (<80%) that could not form bulk materials with structural integrity or be suitable for performance testing. Considering that the boiling point of magnesium (1090 °C) is relatively low, its volatilization increases dramatically under high temperature and the vacuum conditions used in this experiment. To avoid composition and property deviations caused by the loss of principal elements, the maximum sintering temperature was limited to 1000 °C. Therefore, four sintering temperatures—850 °C, 900 °C, 950 °C, and 1000 °C—were selected for this study. Cylindrical samples with a diameter of 20 mm and a height of approximately 3 mm were finally obtained. For ease of reference, the bulk alloys sintered at 850 °C, 900 °C, 950 °C, and 1000 °C were designated as LW850, LW900, LW950, and LW1000, respectively.

The sintered alloy specimens obtained via SPS were sequentially ground on both top and bottom surfaces using SiC abrasive papers from 60 to 2000 grit, followed by mechanical polishing with diamond polishing spray. Subsequently, the specimens were cleaned with deionized water and anhydrous ethanol and then dried under a cold air stream.

The phase composition of the high-entropy alloy powders and the sintered bulk alloys was analyzed using a Panalytical Empyrean X-ray diffractometer (XRD) (PANalytical B.V., Almelo, The Netherlands). The measurement was conducted with an operating voltage of 45 kV and a current of 40 mA, using a scan range of 10° to 90° and a scanning speed of 6°/min. The microstructure and elemental distribution of the alloys were examined using a HITACHI SU8100 field-emission scanning electron microscope (SEM) (Hitachi High-Technologies Corporation, Tokyo, Japan) equipped with an Oxford ULTIM MAX 40 energy dispersive spectrometer (EDS) (Oxford Instruments, Abingdon, UK). The particle size distribution of the alloy powders was measured with a Bettersize 2600E laser particle size analyzer (Dandong Bettersize Instruments Ltd., Dandong, China).

Based on Archimedes' principle, the density of the sintered MgAlTiVFeCo high-entropy alloy samples was measured using an SJ-300G solid densitometer (Shanghai Suju Instrument Technology Co., Ltd., Shanghai, China), with at least three repeated measurements for each sample. The microhardness of the alloys was tested using an HV-1000Z Vickers microhardness (Shanghai Aolong Xingdi Testing Equipment Co., Ltd., Shanghai, China) tester under a load of 4.9 N and a dwell time of 10 s. For each sample,

measurements were performed randomly on at least five different areas, and the average value was taken.

The high-temperature oxidation experiment of the alloy was conducted in a BSK-C25 muffle furnace (Luoyang BSK Electronic Materials Co., Ltd., Luoyang, China). The test alloy samples were cut into small pieces, cleaned, dried, and then weighed using an electronic balance and measured with a vernier caliper to record their dimensions. The oxidation test was performed at 900 °C for 60 h, with samples taken every 5 h. Before and after the oxidation test, the samples and crucibles were weighed using a precision electronic balance with an accuracy of 0.1 mg. The obtained data were analyzed to derive the weight gain curve at 900 °C and to calculate the oxidation rate constant, k_p .

At room temperature, the corrosion resistance of the high-entropy alloy samples was evaluated using an electrochemical workstation (CHI650E) (CHI650E, Shanghai Chenhua Instrument Co., Ltd., Shanghai, China). A platinum plate served as the counter electrode (CE), and a saturated calomel electrode (SCE) was used as the reference electrode. The corrosive medium was a 3.5 wt% NaCl solution. Prior to testing, the sample was immersed in the solution for 30 min to establish a stable open circuit potential (OCP). Electrochemical impedance spectroscopy (EIS) was then performed at the OCP over a frequency range from 10^5 to 10^{-2} Hz with an amplitude of 5 mV. Potentiodynamic polarization (Tafel) curves were measured with a scan range from -1 V to 1 V and a scan rate of $0.01 \text{ V} \cdot \text{s}^{-1}$. An equivalent circuit model was constructed using Zview software (Version 3.3, Scribner Associates Inc., Southern Pines, NC, USA) to fit the impedance data.

3. Results and Discussion

3.1. Phase Composition and Surface Morphology of MgAlTiVFeCo Powder

Figure 1 shows the XRD patterns of the alloy powders after different milling times. As seen in Figure 1, diffraction peaks corresponding to $\text{Al}_{12}\text{Mg}_{17}$ from the MgAl alloy powder, as well as Al, Ti, V, Fe, and Co, are all clearly visible in the unmilled powder. After 20 h of milling, the intensity of all diffraction peaks significantly decreased, and a BCC phase was formed. With increasing milling time, the diffraction peaks of the BCC phase gradually broadened. Finally, after 100 h of milling, all other peaks disappeared, and a single BCC solid solution phase was formed. The corresponding crystallographic plane indices are (110), (200), and (211).

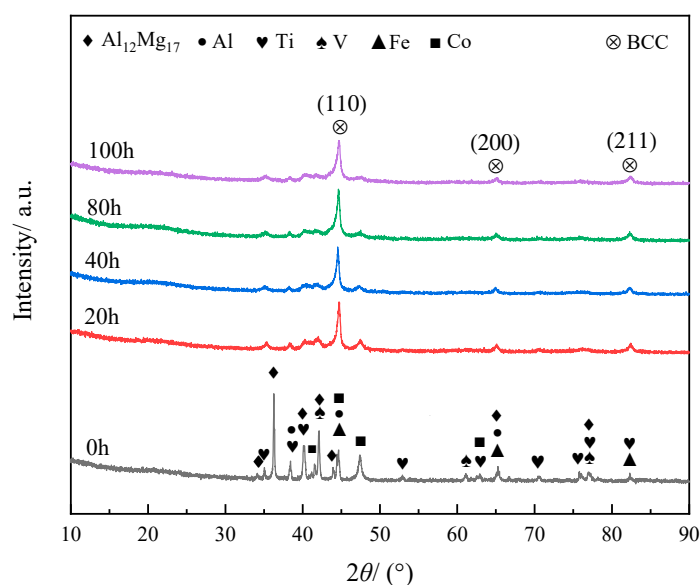


Figure 1. XRD patterns of MgAlTiVFeCo powders after ball milling for various durations (0–100 h). The formation of a single BCC solid solution phase is confirmed after 100 h.

The crystallite size was calculated using Scherrer's formula:

$$D = 0.89 \lambda / (\beta \cos \theta) \quad (1)$$

where D is the crystallite size, λ is the X-ray wavelength, β is the full width at half maximum (FWHM) of the diffraction peak, and 2θ is the diffraction angle. Assuming that microstrain is neglected, the calculated crystallite sizes of the alloy powders are summarized in Table 1. As the milling time increases, the crystallite size gradually decreases. The crystallite size corresponding to the (110) plane of the powder milled for 100 h is 17.66 nm.

Table 1. Parameters of (110) crystal plane diffraction peak of alloy powder with different milling times.

Milling Time/h	$2\theta/(^{\circ})$	FWHM/ $(^{\circ})$	Lattice Constant/nm	Grain Size/nm
20	44.62	0.428	0.28690	19.82
40	44.47	0.439	0.28782	19.33
80	44.56	0.457	0.28727	18.58
100	44.62	0.481	0.28690	17.66

It should be noted that the Scherrer formula does not account for the contribution of microstrain to diffraction peak broadening. Given that the high-energy ball milling process introduces a substantial number of crystal defects, the Williamson–Hall method was further employed to analyze the diffraction data of the powders ball-milled for 100 h. This method is based on the following equation:

$$\beta \cos \theta = 0.89 \lambda / D + 4\epsilon \sin \theta \quad (2)$$

where ϵ represents the microstrain.

The results indicate that both significant grain refinement and microstrain coexist in the powders, with an estimated microstrain value of approximately 0.38%. After separating the strain contribution, the corrected crystallite size is about 22 nm. The values derived from the Scherrer formula in Table 1 can be regarded as lower-bound estimates of the crystallite size, whereas the evolutionary trend that “the crystallite size decreases monotonically with ball-milling time” remains valid regardless of the analytical model used.

Figure 2 shows the microscopic morphology of the alloy powders after different ball-milling durations. With increasing milling time, significant changes in the morphology, size, and uniformity of the powders are observed. After 20 h of milling (Figure 2a), the powder particles are relatively large, exhibit irregular shapes with numerous edges and flaky structures, indicating that the powder is still in the initial stage of fracture and plastic deformation. After 40 h of milling (Figure 2b), the particle size decreases noticeably, edges become rounded, and some particles undergo cold-welding and agglomeration, leading to a more uniform morphology. Upon further milling to 80 h (Figure 2c), the powder is further refined, the particle-size distribution becomes narrower, the shapes approach equiaxed, and agglomeration is somewhat alleviated, reflecting the continuous energy accumulation and structural evolution during the milling process. When the milling time is extended to 100 h (Figure 2d), the powder exhibits a more homogeneous fine-particle morphology with clear particle boundaries and reduced agglomeration, demonstrating that prolonged milling promotes sufficient refinement and homogenization of the powder, which is beneficial for subsequent densification and the formation of alloy phases [20].

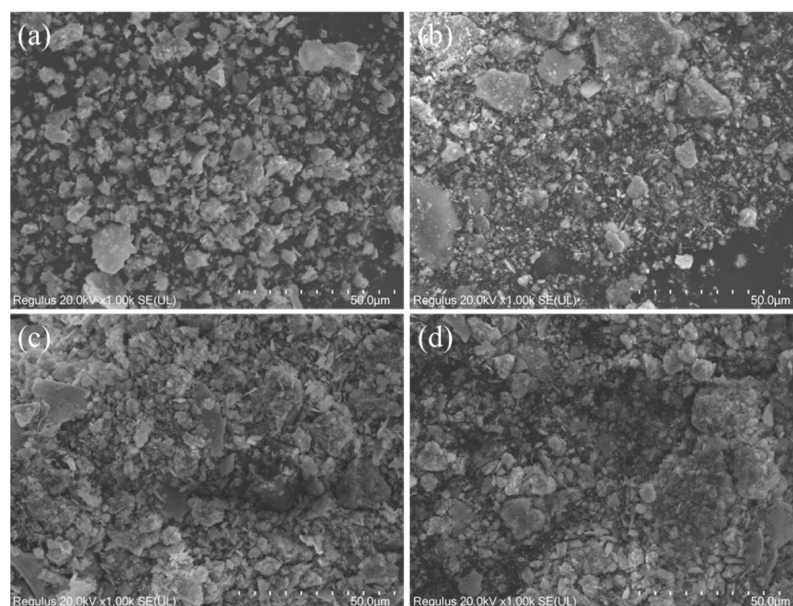


Figure 2. SEM images showing the evolution of powder morphology with increasing ball milling time (20–100 h). The particles transition from irregular flakes to finer, more equiaxed shapes: (a) 20 h; (b) 40 h; (c) 80 h; (d) 100 h.

Figure 3 shows the particle-size distribution and elemental mapping of the alloy powders ball-milled for 100 h. The particle-size distribution histogram reveals a relatively broad unimodal profile, with the main distribution ranging from 0.1 μm to 45 μm , along with a small fraction of coarse particles exceeding 50 μm . This distribution indicates that after prolonged milling for 100 h, a dynamic equilibrium has been established between powder refinement and particle cold-welding/agglomeration. The median particle size ($D_{50} = 11.10 \mu\text{m}$) is expected to be significantly reduced compared to the initial mixed powders, yet complete nanoscale refinement was not achieved. This can be attributed to the high work-hardening rate of the high-entropy alloy system, as well as the coexistence of ductile components (e.g., Al, Fe) and hard phases (e.g., oxides formed by O with Mg and V introduced during long-term milling). These factors enhance the cold-welding tendency of particles after reaching a certain size, thereby hindering further homogeneous refinement. The elemental mapping displays a uniform distribution of all components, demonstrating that after 100 h of high-energy ball milling, thorough mechanical alloying has been achieved, with extensive inter-diffusion among the elements, leading to the preliminary formation of a compositionally homogeneous solid-solution precursor.

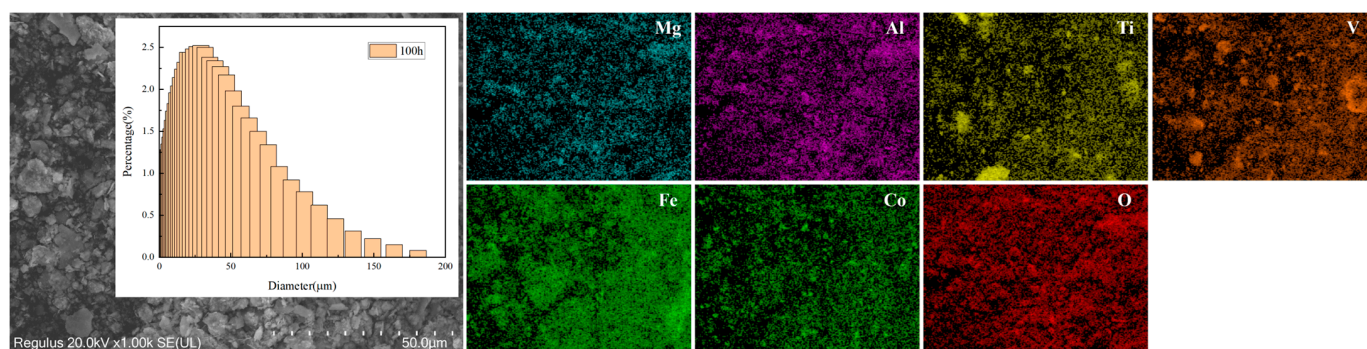


Figure 3. Particle size distribution and elemental maps of powder after 100 h of ball milling. The powder exhibits a D_{50} of 11.10 μm and a uniform distribution of all constituent elements.

To analyze the evolution of oxygen introduced during mechanical alloying in the subsequent sintering process, energy-dispersive X-ray spectroscopy (EDS) was employed to perform statistical point analysis of the oxygen content in the 100 h ball-milled powder and the bulk samples sintered at different temperatures. The average oxygen content (in atomic percent, at%) for each sample was calculated based on multiple independent measurements, as presented in Table 2. It should be noted that EDS provides semi-quantitative data for light elements such as oxygen. Nevertheless, compared with the as-milled powder, the average bulk oxygen content of all sintered samples decreased significantly, indicating redistribution or partial loss of oxygen during sintering. Within the sintering temperature range of 850 °C to 1000 °C, the oxygen content of the samples did not exhibit a simple monotonic variation. This non-monotonic trend suggests that sintering temperature not only influences the diffusion and reaction rates of oxygen but may also complexly affect the existing forms and distribution of oxygen by altering the densification behavior and phase composition of the alloy.

Table 2. Average oxygen content of ball-milled powder and sintered MgAlTiVFeCo alloy (EDS analysis).

Sample	Average O Content/(at%)	Number of Measurements
100 h Powder	50.97	3
LW850	31.30	5
LW900	36.34	5
LW950	30.66	5
LW1000	27.16	5

3.2. Phase Composition of the MgAlTiVFeCo High-Entropy Alloy

Figure 4 shows the XRD patterns of the alloy powder milled for 100 h after sintering at different temperatures. As shown in Figure 4, the position and profile of the BCC phase diffraction peaks remain consistent across samples sintered at different temperatures. However, for the LW1000 sample, the intensity of the main BCC phase diffraction peak increases and the peaks become sharper, indicating an increase in grain size and more complete crystallinity at the 1000 °C sintering temperature. Compared to the phase structure of the as-milled powder, the sintered samples exhibit the precipitation of secondary phases. This demonstrates that the nanocrystalline precursor powder formed via mechanical alloying successfully crystallized during the rapid SPS process, resulting in a high-entropy alloy solid solution with a BCC matrix. By comparison with standard PDF cards, the secondary phases are identified as various spinel-type oxides (MgV_2O_4 (PDF#65-3109), MgAl_2O_4 (PDF#82-2424), FeAl_2O_4 (PDF#82-0584), AlV_2O_4 (PDF#77-2131)), oxides (V_3O_4 (PDF#34-0615), TiO (PDF#08-0117)), and an ilmenite-type oxide (FeTiO_3 (PDF#75-0519)). The presence of oxides in the alloy can be attributed to two main factors. First, during the weighing process, the raw materials inevitably came into contact with atmospheric oxygen, leading to the formation of a thin oxide film on the surface of some elemental powders. Second, during high-energy ball milling, the powder particles underwent severe plastic deformation and friction both on the surface and internally. This accelerated elemental diffusion, providing the driving force for oxide precipitation, while the defects and stresses generated during mechanical alloying supplied nucleation sites, thereby promoting oxide formation [21].

Studies have shown that magnesium vanadium spinel (MgV_2O_4) and magnesium aluminate spinel (MgAl_2O_4) can form at relatively low temperatures, whereas spinels such as AlV_2O_4 and FeAl_2O_4 , as well as FeTiO_3 , require higher temperatures to precipitate stably. As shown in Figure 4, with increasing sintering temperature, the diffraction peak intensities of the MgV_2O_4 and MgAl_2O_4 spinel phases increase significantly. This indicates that

at lower temperatures, sintering primarily activates short-range diffusion at the powder surfaces, allowing the pre-formed Mg, Al, and V oxides from the milling process to undergo complex reactions, crystallize, and grow. However, when the sintering temperature rises to 1000 °C, phases such as AlV_2O_4 , FeAl_2O_4 , and FeTiO_3 precipitate. This is attributed to the fact that, within the severe lattice distortion characteristic of high-entropy alloys, sufficiently high temperature and sintering pressure provide atoms with extremely high diffusion activity, promoting oxide growth and elemental redistribution. V_3O_4 is an intermediate-valence oxide, and its formation is closely related to the multivalent nature of vanadium and the local oxygen partial pressure. During the SPS process, trace oxygen adsorbed on powder surfaces or in inter-particle gaps preferentially reacts with the more reactive V. Due to V's strong affinity for oxygen and its relatively fast diffusion rate, V tends to accumulate at grain boundaries or defects to form V_3O_4 . In contrast, Ti diffuses more slowly in the alloy and is prone to competing for oxygen with more reactive elements such as Al and V; hence, the oxidation of Ti often remains in a low-valence state rather than being fully oxidized to TiO_2 . When the sintering temperature reaches 1000 °C, the TiO phase disappears in the LW1000 sample, suggesting that at sufficiently high temperatures, the extreme atomic mobility may temporarily overcome diffusion barriers, causing partial dissolution of oxides and driving the system metastably toward a more homogeneous single-phase solid solution. Therefore, the phase evolution in the alloy samples is the result of a competition between thermodynamic driving forces and kinetic diffusion rates under the rapid sintering conditions of SPS. Thermodynamically, elements such as Mg, Al, Ti, and V have a very strong affinity for oxygen, making oxide formation energetically favorable. Kinetically, the rapid heating/cooling characteristics of the SPS process, combined with the sluggish diffusion kinetics (the “sluggish diffusion effect”) typical of high-entropy alloys, significantly influence the final outcome of this process.

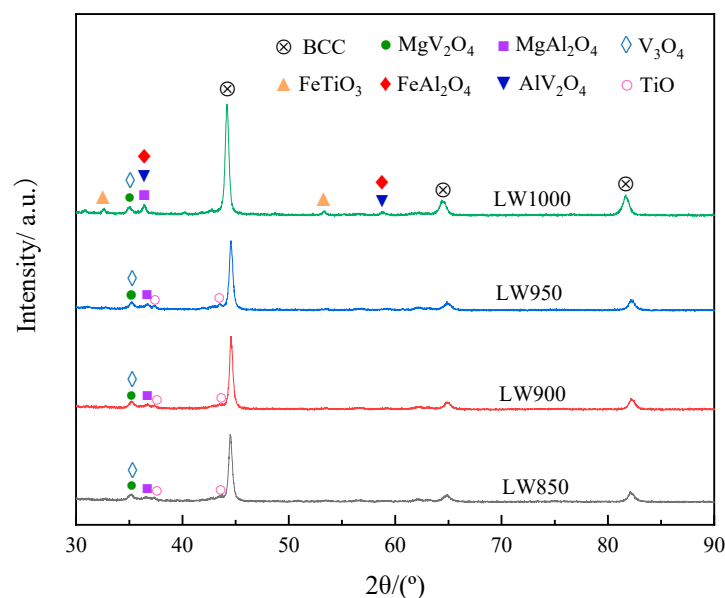


Figure 4. XRD patterns of the 100 h milled powder after SPS at different temperatures (850–1000 °C). The BCC matrix is retained, and various secondary oxide phases precipitate, with their intensity increasing with temperature.

3.3. Microstructure of the MgAlTiVFeCo High-Entropy Alloy

Figure 5 shows the SEM images of the alloys sintered at different temperatures. As shown in Figure 5, the surface of the LW850 (850 °C) alloy exhibits a relatively loose structure with noticeable pores. Compared to LW850, the LW900 (900 °C) alloy shows a slight reduction in porosity and gradually blurred particle boundaries, but the overall

morphology remains inhomogeneous. The densification of the LW950 (950 °C) alloy is significantly improved, with a marked decrease in porosity and the appearance of locally dense regions, although a small amount of finely dispersed precipitated phases is still visible. The surface of the LW1000 (1000 °C) alloy is further densified, forming a continuous matrix with a more uniform distribution of precipitated phases.

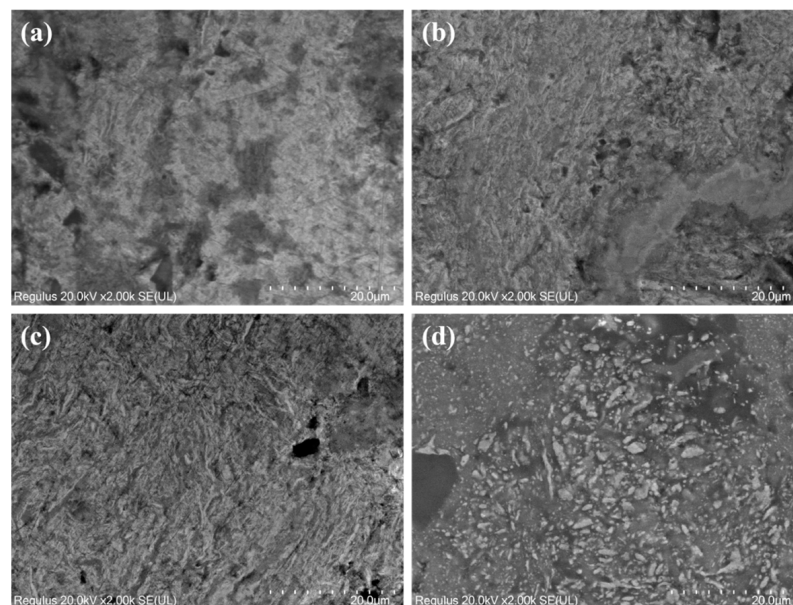


Figure 5. BSE images of alloys sintered at different temperatures. Microstructural densification improves significantly with increasing sintering temperature, from a porous structure (LW850) to a dense, continuous matrix (LW1000): (a) LW850, (b) LW900, (c) LW950, (d) LW1000.

To establish a clear correspondence among the microscopic morphology, local chemical composition, and overall phase constitution, a multi-scale correlation analysis was employed. First, based on the atomic-number contrast in back-scattered electron (BSE) images, typical regions exhibiting distinct gray-level intensities (such as Regions 1, 2, and 3 illustrated in Figure 6) were identified. Subsequently, high-resolution energy-dispersive X-ray spectroscopy (EDS) point analyses and small-area elemental mapping were performed on these specific regions to obtain their precise chemical compositions (Table 3). Finally, these local composition data were systematically compared and correlated with the crystal structure information determined by large-area X-ray diffraction (XRD).

Table 3. EDS results (at%) of alloys sintered at different temperatures (Figure 5).

Alloy	Zone	Mg	Al	Ti	V	Fe	Co	O	C
LW850	1	3.1	2.8	7.0	11.6	24.6	10.9	28.0	11.9
	2	7.4	15.5	3.2	2.0	3.7	1.0	40.1	11.1
	3	0.1	0.2	40.5	1.0	4.8	1.8	24.2	27.4
LW900	1	4.4	6.0	8.3	7.6	28.9	11.6	33.2	--
	2	7.8	11.3	11.2	6.1	9.9	2.5	51.2	--
	3	0.8	2.8	37.2	4.9	10.5	5.0	38.6	--
LW950	1	3.1	3.5	8.0	9.6	34.6	12.0	29.3	--
	2	4.3	5.0	4.0	10.2	35.3	14.3	26.9	--
	3	2.9	2.4	3.0	35.2	21.9	7.8	30.6	--
LW1000	1	4.2	6.4	1.4	3.1	40.1	33.2	11.7	--
	2	12.8	26.8	1.3	1.8	1.9	0.4	55.0	--
	3	9.9	7.2	9.9	16.5	6.6	2.1	47.7	--

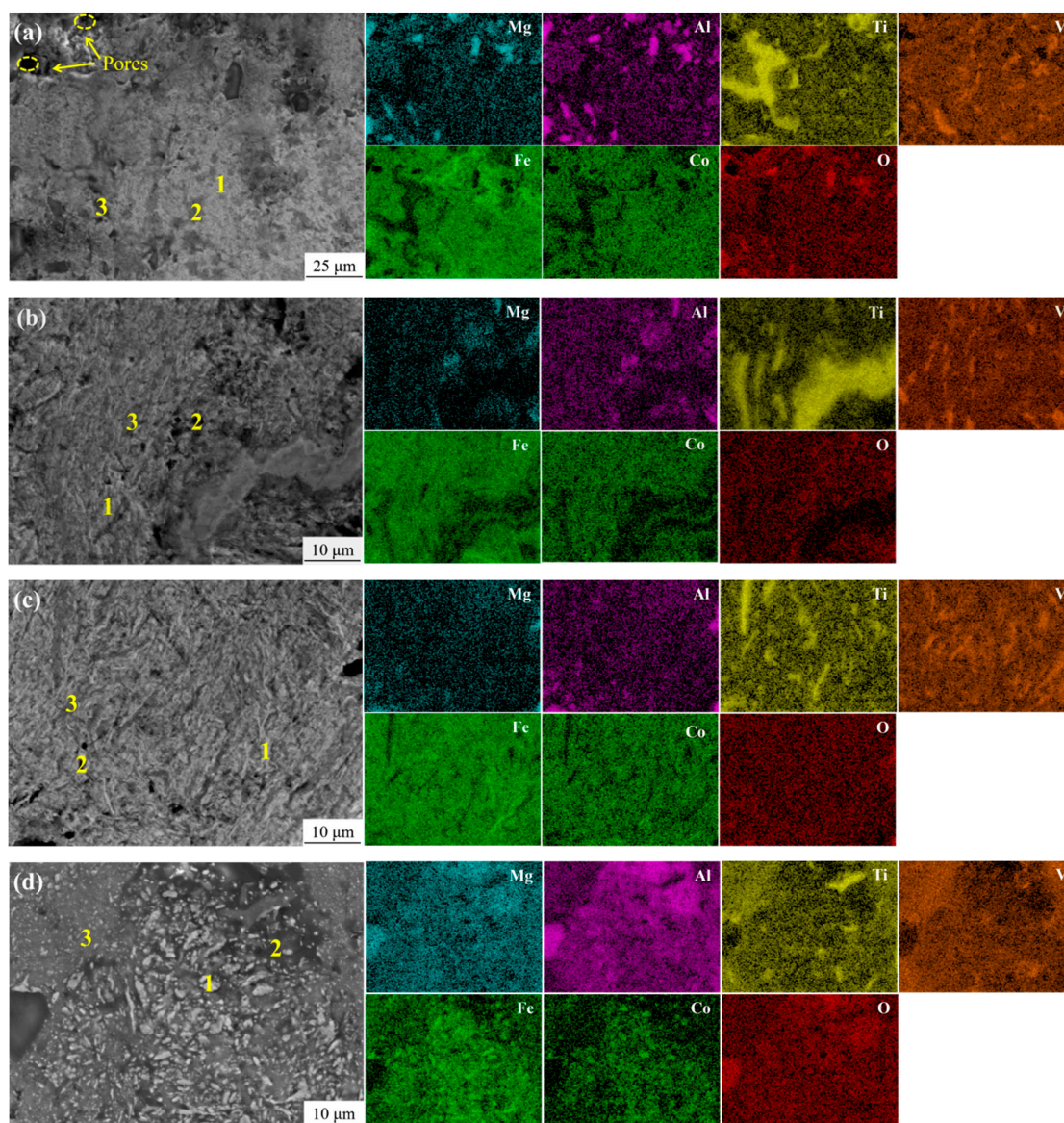


Figure 6. Elemental mapping (Mg, Al, Ti, V, Fe, Co) on the surfaces of alloys sintered at different temperatures: (a) LW850, (b) LW900, (c) LW950, (d) LW1000.

In the BSE images, regions displaying light-gray contrast (labeled as Region 1) show a relatively uniform distribution of the principal elements (Mg, Al, Ti, V, Fe, Co) according to EDS, with typically elevated contents of Fe and Co (Table 3). The compositional characteristics of this region are consistent with the dominant body-centered cubic (BCC) solid-solution phase identified in the XRD patterns. The ability of the BCC structure to accommodate elements with considerably different atomic radii, together with the lattice parameters calculated from the diffraction peak positions (Figure 4), which lie between those of the major constituent elements, further supports that this phase is a multi-principal-element substitutional solid solution rather than a single intermetallic compound. Regions exhibiting dark-gray contrast (labeled as Region 2) show strongly overlapping and intense signals of Mg, Al, and O in their EDS spectra. For example, in Region 2 of sample LW1000, the combined atomic percentages of Mg, Al, and O exceed 90%. This Mg-, Al-, and O-rich composition perfectly matches the diffraction peaks of various spinel-type structures (AB_2O_4), such as $MgAl_2O_4$ and MgV_2O_4 , detected by XRD. The spinel phase allows a range of divalent and trivalent cations (e.g., Mg^{2+} , Fe^{2+} , Al^{3+} , V^{3+}) to occupy its tetrahedral (A) and octahedral (B) sites, leading to slight variations in the cation ratios measured by EDS

across different samples, which explains the minor compositional differences observed in Region 2 of various specimens. Regions with a deep-gray contrast (labeled as Region 3) are identified by EDS analysis as Ti-rich (or V-rich) areas associated with high oxygen content. This compositional feature corresponds clearly to the diffraction peaks of simple oxide phases detected by XRD, such as TiO (PDF#08-0117) or V₃O₄ (PDF#34-0615). The precipitation of these phases is attributed to local micro-heterogeneities in oxygen partial pressure and elemental diffusion kinetics.

In multi-component alloy systems, the spinel structure (AB₂O₄) possesses higher configurational entropy and structural stability. During the SPS process, the alloy remains under relatively low-oxygen (or locally oxygen-deficient) conditions. According to the Ellingham diagram, under a given oxygen partial pressure, the oxidation tendency of the metals follows the order: Mg > Al > Ti > V > Fe > Co. However, in the MgAlTiVFeCo high-entropy alloy, although Mg and Al have strong oxidation tendencies, they are more likely to combine with other trivalent metals (e.g., Al³⁺, Fe³⁺, V³⁺) to form spinels rather than individually forming MgO or Al₂O₃. In contrast, the oxidation tendencies of Fe and Co are relatively weak. In the limited oxygen environment, oxygen is preferentially consumed by Mg, Al, Ti, and V, thereby suppressing the formation of FeO and CoO. This indicates that during the sintering of multi-component lightweight high-entropy alloys, trace oxygen can lead to the formation of complex oxide phases through selective oxidation, which in turn influences the microstructure and properties of the alloy.

3.4. Physical Properties of the MgAlTiVFeCo High-Entropy Alloy

According to the rule of mixtures based on the density of each constituent element, the theoretical density of the alloy can be calculated using Equation (3):

$$\rho_{mix} = \frac{\sum_{i=1}^n c_i A_i}{\sum_{i=1}^n c_i A_i / \rho_i} \quad (3)$$

where A_i is the atomic weight of element i , c_i is its atomic concentration, and ρ_i is the density of element i . The calculated theoretical density of the MgAlTiVFeCo high-entropy alloy is 5.304 g/cm³, which meets the design criterion for lightweight high-entropy alloys (density < 6 g/cm³). The measured actual densities of the LW850, LW900, LW950, and LW1000 alloy samples are 4.165 g/cm³, 4.553 g/cm³, 4.641 g/cm³, and 5.127 g/cm³ (Figure 7a), respectively. The corresponding relative densities, calculated from these values, are 85.8%, 87.5%, 89.3%, and 96.7%.

Figure 7b shows the microhardness and relative density of the alloys sintered at different temperatures. As illustrated in Figure 7b, the variation trends of microhardness and relative density for the alloy samples are consistent, both reaching their peak values at the sintering temperature of 1000 °C. The maximum hardness of LW1000 is 1235.5 HV, which is significantly higher than that of reported high-entropy alloys such as AlCrCuFeTi (714 HV) and AlCrCuFeV (565 HV) [22], as well as traditional hard alloys [23]. The relative density of this alloy reaches 96.7%. LW1000 thus exhibits excellent physical properties characterized by high hardness and high density.

The sintering densification process is closely related to the evolution of pore structure. At lower sintering temperatures (850 °C and 900 °C), material transport is dominated by surface diffusion, and atomic mobility is limited. Consequently, the interconnected pore channels formed between the initial powder particles are difficult to eliminate, leading to slow densification progress. The residual porosity primarily consists of open pores, which can serve as rapid pathways for corrosive media or oxygen penetration. As the temperature rises to 950 °C or above, bulk diffusion and grain-boundary diffusion mechanisms become activated. The sintering driving force is significantly enhanced, accelerating atomic

migration and neck growth between particles, which promotes gradual spheroidization, shrinkage, and closure of pores. As a result, the pore structure transitions to one dominated by isolated closed pores. At 1000 °C, this process proceeds most completely, thereby yielding the highest relative density (96.7%).

As can be seen from Figure 7b, during low- to medium-temperature sintering (850–950 °C), the alloy hardness increases only gradually with rising temperature, primarily due to reduced porosity and improved matrix continuity. At the high sintering temperature of 1000 °C, the alloy's hardness increases substantially. Besides the further improved densification, this significant enhancement is closely related to the abundant precipitation of oxide secondary phases. Spinel-type oxides (e.g., MgAl_2O_4 , FeAl_2O_4) possess high hardness and high modulus; their dispersed distribution can effectively hinder dislocation motion. Simple oxides such as V_2O_5 and TiO_2 , as well as the ilmenite-type oxide FeTiO_3 , precipitate at grain boundaries or phase boundaries, pinning grain boundaries and refining grains, thereby contributing to precipitation strengthening and grain-refinement strengthening. LW1000 thus exhibits physical properties characterized by high hardness and high densification.

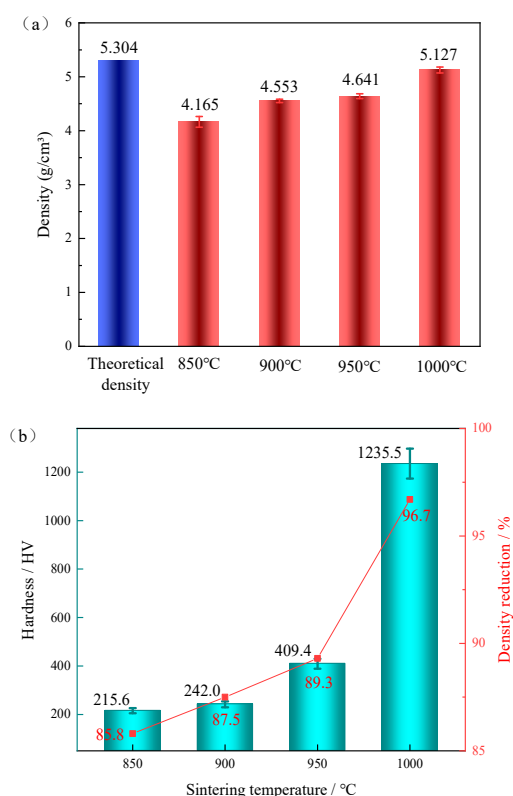


Figure 7. (a) Theoretical and actual density and (b) microhardness and relative density of alloys sintered at different temperatures. Both hardness and densification peak at 1000 °C (LW1000: 1235.5 HV, 96.7%).

3.5. High-Temperature Oxidation Resistance of the MgAlTiVFeCo High-Entropy Alloy

Figure 8 shows the oxidation kinetics curves of the MgAlTiVFeCo high-entropy alloy oxidized at 900 °C for 60 h. According to the weight gain per unit area curves in Figure 8a, all samples follow a parabolic law during the 60 h oxidation at 900 °C, which is consistent with Wagner's theory of high-temperature oxidation [24]. This indicates that the oxidation process is solid-state diffusion-controlled. The oxidation weight gain can be divided into two stages: the first stage (0–25 h), where the oxidation rate is relatively fast, and the weight gain increases rapidly with time, corresponding to the initial formation and growth of

the oxide scale, and the second stage (25–60 h), where the oxidation rate slows down and stabilizes, corresponding to the thickening of the oxide scale and the establishment of a diffusion equilibrium. After 60 h of oxidation, the weight gains for LW850, LW900, LW950, LW1000, and LW1050 are 290.43, 284.08, 301.46, 130.76, and 140.45 mg/cm², respectively.

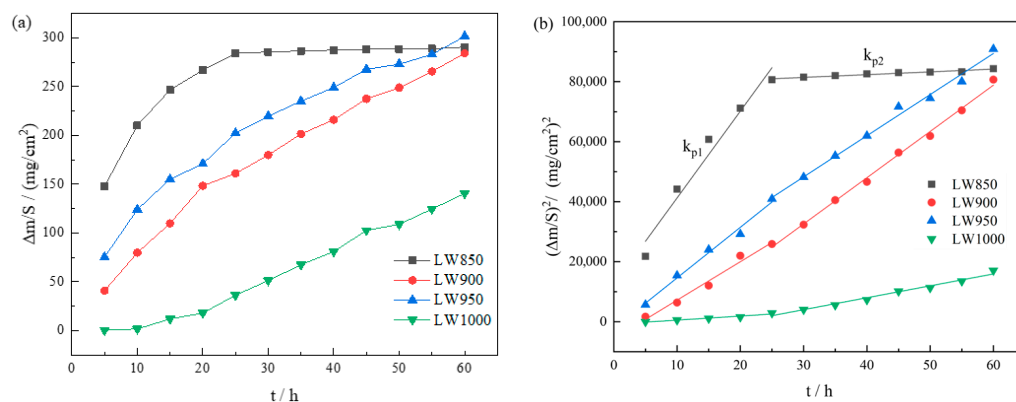


Figure 8. (a) Weight gain curves and (b) parabolic plots for oxidation at 900 °C for 60 h. All samples follow parabolic kinetics, with LW1000 exhibiting the lowest oxidation rate constants in both stages.

The parabolic rate constant (k_p) can be calculated using Equation (4):

$$\left(\frac{\Delta m}{S}\right)^2 = k_p t \quad (4)$$

where Δm is the mass gain, S is the surface area of the sample, and t is the oxidation time.

Figure 8b shows the relationship between $(\Delta m/S)^2$ and t or the MgAlTiVFeCo high-entropy alloy. The obtained oxidation rate constants (k_{p1} , k_{p2}) and the corresponding goodness-of-fit coefficients (R^2) are listed in Table 4. As can be seen from the figure and Table 4, during the first oxidation stage (0–25 h), the order of k_{p1} values is $k_{p1}(\text{LW1000}) < k_{p1}(\text{LW900}) < k_{p1}(\text{LW950}) < k_{p1}(\text{LW850})$. In the second oxidation stage (25–60 h), the order is $k_{p2}(\text{LW850}) < k_{p2}(\text{LW1000}) < k_{p2}(\text{LW950}) < k_{p2}(\text{LW900})$. The k_p values exhibit a non-monotonic variation with increasing sintering temperature, reflecting that the sintering temperature influences the oxidation diffusion path and rate by altering the intrinsic structure of the alloy. The oxidation rate constant of the MgAlTiVFeCo high-entropy alloy is comparable to that of the CrNbTiVAl_x high-entropy alloy reported by Zhu et al. [25] (Table 4), indicating that the alloy exhibits favorable high-temperature oxidation resistance at 900 °C.

Table 4. High temperature oxidation rate constants of MgAlTiVFeCo high-entropy alloy at 900 °C.

Alloy	t/h	$k_{p1}/(\text{mg}^2 \cdot \text{cm}^{-4} \cdot \text{s}^{-1})$	R^2	t/h	$k_{p2}/(\text{mg}^2 \cdot \text{cm}^{-4} \cdot \text{s}^{-1})$	R^2	Ref.
LW850	0–25 h	8.04×10^{-1}	0.96	25–60 h	2.56×10^{-2}	0.95	This work
LW900 °C	0–25 h	3.56×10^{-1}	0.98	25–60 h	4.28×10^{-1}	0.99	
LW950 °C	0–25 h	4.68×10^{-1}	0.99	25–60 h	3.81×10^{-1}	0.99	
LW1000 °C	0–25 h	3.76×10^{-2}	0.95	25–60 h	1.10×10^{-1}	0.98	
CrNbTiVAl _{0.25}	0–40 h	1.98×10^{-1}	0.94	40–100 h	2.90×10^{-2}	0.97	[25]
CrNbTiVAl _{0.5}	0–40 h	1.43×10^{-1}	0.96	40–100 h	5.31×10^{-2}	0.97	
CrNbTiVAl _{0.75}	0–40 h	8.81×10^{-2}	0.93	40–100 h	7.23×10^{-2}	0.98	
CrNbTiVAl	0–40 h	5.71×10^{-2}	0.94	40–100 h	8.91×10^{-2}	0.99	

The relative density of the alloy increases significantly with the sintering temperature. The LW850 sample, containing numerous pores and interconnected channels, provides fast pathways for oxygen in-diffusion during the initial stage of high-temperature oxidation, accelerating internal oxidation and resulting in a higher k_p value in the first stage. Conversely, high-density alloys like LW1000, with fewer pores and tightly bonded grain boundaries, effectively impede oxygen diffusion into the interior, lowering the initial oxidation rate. The various secondary oxide phases (spinel-type, simple oxides, ilmenite-type) formed during sintering play different roles during high-temperature oxidation. Spinel-type oxides (e.g., $MgAl_2O_4$, $FeAl_2O_4$), possessing excellent structural stability and low oxygen diffusion coefficients, are uniformly distributed in the LW1000 microstructure. During oxidation, they act as diffusion barriers, inhibiting further oxygen penetration. The content of non-protective oxides like V_3O_4 and TiO gradually decreases with increasing sintering temperature, reducing their detrimental effect on the continuity of the oxide scale. When ilmenite-type oxides such as $FeTiO_3$ coexist with spinel phases, they can form multi-layered or composite oxide scales, improving the scale's densification and adherence, thereby further enhancing the high-temperature oxidation resistance of LW1000. The high-temperature oxidation process is controlled by grain boundary diffusion. Alloys sintered at lower temperatures have finer grains and a higher density of grain boundaries, providing rapid pathways for oxygen diffusion along the boundaries, which is detrimental to oxidation resistance. In contrast, alloys sintered at high temperatures exhibit grain growth and a reduced density of grain boundaries. Furthermore, protective oxides may become enriched at these boundaries, hindering oxygen diffusion and thus leading to a lower oxidation rate.

Figure 9 shows the surface oxide-layer morphology and the corresponding elemental mapping of the $MgAlTiVFeCo$ high-entropy alloy after oxidation at 900 °C for 60 h. As the sintering temperature increases, the surface morphology after oxidation exhibits significant systematic changes. The oxide layer on the LW850 sample (Figure 9a) appears rough, with a loose accumulation of granular oxides and noticeable micro-cracks and pores. This porous and poorly coherent structure provides fast diffusion channels for oxygen inward transport, which is detrimental to oxidation resistance. The surface compactness of the LW900 sample (Figure 9b) improves to some extent, but the oxide particles remain non-uniform in size, and defects with poor continuity are still present in local areas. When the sintering temperature rises to 950 °C (Figure 9c) and 1000 °C (Figure 9d), a fundamental transformation occurs in the oxide-layer morphology. The surfaces become relatively smooth and dense, forming continuous and well-covered oxide layers. In particular, the LW1000 sample displays the most uniform and compact surface morphology, with micro-cracks and pores nearly eliminated. Such a dense oxide layer effectively hinders further inward diffusion of oxygen into the substrate, representing a key microstructural feature for excellent oxidation resistance.

The EDS elemental mapping results (elemental distribution maps in Figure 9a–d) further clarify the chemical composition and formation mechanism of the oxide layer. In all specimens, oxygen (O) exhibits a uniform planar distribution that closely coincides with the oxide-layer regions, confirming the extensive formation of surface oxides. Magnesium (Mg) and aluminum (Al) are significantly enriched in the oxide layer, particularly in the LW950 and LW1000 specimens, where their enriched regions are more continuous. This is attributed to the high affinity of Mg and Al for oxygen (more negative standard Gibbs free energy, ΔG^θ), which drives their preferential outward diffusion during high-temperature oxidation, leading to the formation of MgO , Al_2O_3 , or their composite oxides on the surface.

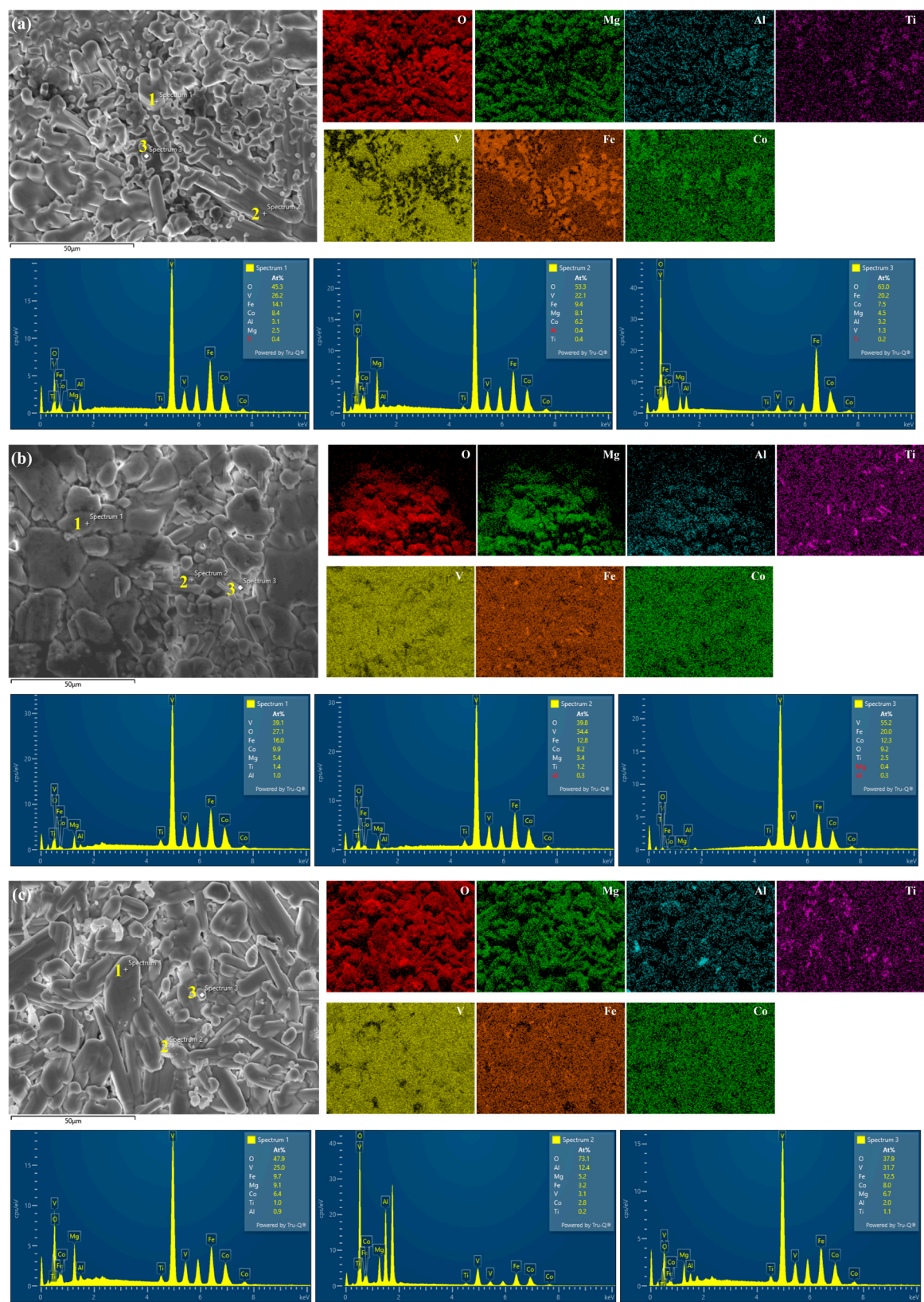


Figure 9. Cont.

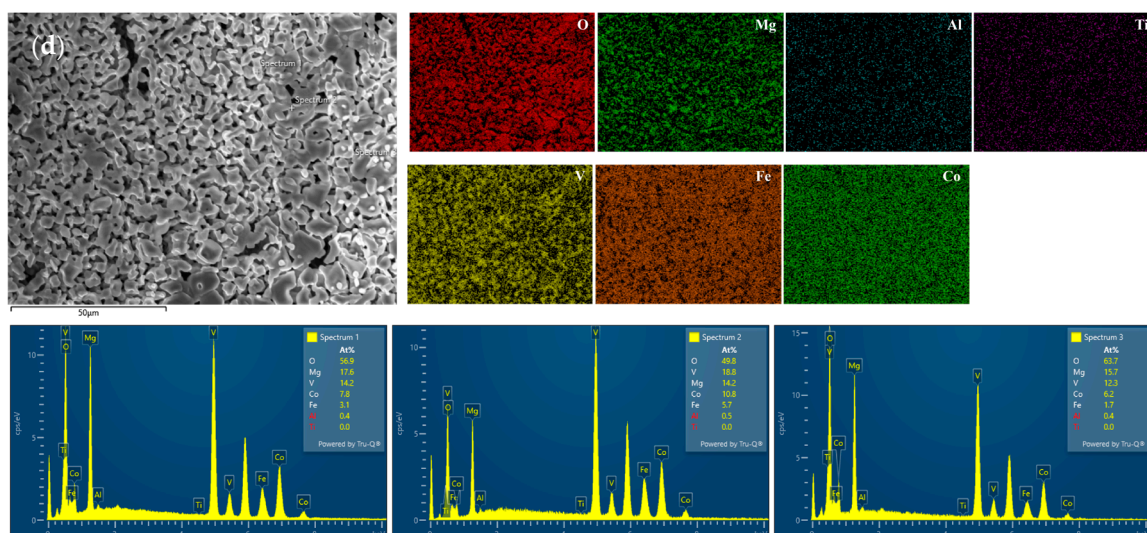


Figure 9. Surface morphology and elemental maps after oxidation at 900 °C for 60 h. The oxide layer transforms from loose and porous (LW850) to dense and continuous (LW1000): (a) LW850, (b) LW900, (c) LW950, (d) LW1000.

The Cross-sectional morphology of specimens sintered at different temperatures (LW850, LW900, LW950, LW1000) after oxidation at 900 °C for 60 h is shown in Figure 10. In all four specimens, the oxide layer is clearly distinguishable and resides on top of the dense alloy substrate. The oxide layer of the LW850 specimen (Figure 10a) is the widest, exhibiting a thick and non-uniform multi-layer structure. Within the oxide layer, numerous longitudinal cracks and pores are observed, and the interface with the substrate is undulating. Local areas show extensive spallation of the oxide layer, indicating poor adhesion. The oxide layer of the LW900 specimen (Figure 10b) is somewhat thinner, but still displays a distinct continuous oxide layer along with internal defects. In the LW950 specimen (Figure 10c), the oxide layer becomes relatively thin and continuous, with reduced fragmentation and a flatter interface with the substrate. In contrast, the oxide-layer region of the LW1000 specimen (Figure 10d) is the narrowest and thinnest among the four. Although micro-cracks were introduced during sample preparation, the overall continuity of the oxide layer remains good, with strong adhesion to the substrate and a sharp, planar interface.

The thick oxide layers observed in LW850 and LW900 originate from their porous and defect-rich initial matrix structures. During oxidation, oxygen readily penetrates into such matrices, leading to intense internal oxidation and the formation of thick, loose oxide scales. These oxide scales possess high internal stress and brittleness, making them prone to cracking and spallation under thermal stress or mechanical stress during sample preparation, which is consistent with the rough and porous morphologies observed on their surfaces. Repeated spallation of the oxide layer exposes fresh substrate, resulting in oxidation kinetics that follow an approximately linear, rapid rate. In contrast, for the LW950 and LW1000 specimens, the high-density matrix promotes predominantly external oxidation. Elemental diffusion leads to the formation of a thin, dense, and protective oxide layer on the surface, primarily consisting of spinel phases such as $(\text{Mg,Fe})(\text{Al,V})_2\text{O}_4$. Although these spinel phases are inherently brittle and micro-cracks may still appear during cross-sectional sample preparation, the oxide layer itself is thinner, denser, and exhibits relatively lower internal stress, thereby maintaining better overall integrity. In particular, the thinnest oxide layer of LW1000 provides direct evidence of its superior oxidation resistance, as the oxidation reaction is effectively confined to a near-surface region.

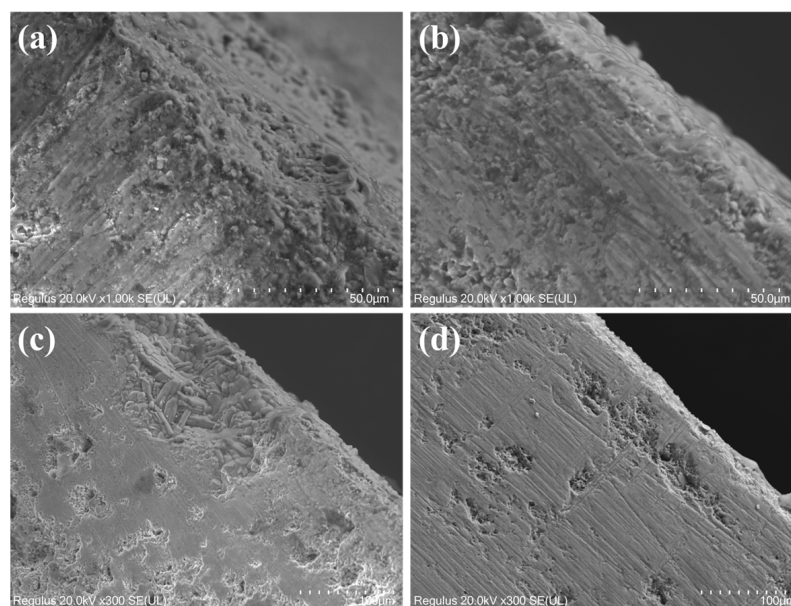


Figure 10. Cross-sectional BSE images of the oxidized samples. The oxide layer thickness decreases with the original sintering temperature, with LW1000 showing the thinnest and most adherent layer: (a) LW850, (b) LW900, (c) LW950, (d) LW1000.

3.6. Electrochemical Corrosion Behavior of the MgAlTiVFeCo High-Entropy Alloy

Figure 11 presents the potentiodynamic polarization curves and electrochemical impedance spectra (EIS) of the MgAlTiVFeCo high-entropy alloys sintered at different temperatures in a 3.5 wt% NaCl solution. From the polarization curves in Figure 11a, it can be observed that LW850, LW900, and LW950 all exhibit active dissolution behavior in the 3.5 wt% NaCl solution without a distinct passivation region. In contrast, LW1000 shows clear active and passivation characteristics, indicating the formation of a dense passive film on its surface, which reduces the anodic dissolution rate of the alloy. The corrosion potential (E_{corr}) and corrosion current density (I_{corr}) obtained by Tafel extrapolation are listed in Table 5. A higher corrosion current density corresponds to a faster corrosion rate and thus poorer corrosion resistance. The corrosion potential (E_{corr}) reflects the tendency of the alloy to corrode; a more negative value indicates a greater susceptibility to corrosion. As shown in Table 5, compared to the alloys sintered at medium-low temperatures (850–950 °C), namely LW850, LW900, and LW950, the high-temperature (1000 °C) sintered alloy LW1000 possesses a more positive corrosion potential (−0.438 V) and a lower corrosion current density ($1.07 \times 10^{-6} \text{ A} \cdot \text{cm}^{-2}$). Through comparison with the electrochemical parameters of lightweight high-entropy alloys reported in the literature [26–29], it can be concluded that the high-temperature sintered LW1000 exhibits excellent corrosion resistance in a 3.5 wt% NaCl solution.

Table 5. Electrochemical performance data of MgAlTiVFeCo high-entropy alloy polarization curve.

Alloy	E_{corr}/V	$I_{\text{corr}}/\text{A} \cdot \text{cm}^{-2}$	Ref
LW850	−0.598	3.67×10^{-4}	This work
LW900	−0.680	1.44×10^{-4}	
LW950	−0.627	5.27×10^{-5}	
LW1000	−0.438	1.07×10^{-6}	
(MgAl) ₇₀ Zn ₂₀ (BiSn) ₁₀	−1.780	6.90×10^{-4}	[26]
AlCuMgMnZn	−1.149	6.89×10^{-6}	[27]
MgTiAlFeNiCr	−0.589	8.914×10^{-5}	[28]
Al ₂ Ti ₃ V ₂ ZrNb (1300 °C)	0.084	8.90×10^{-6}	[29]

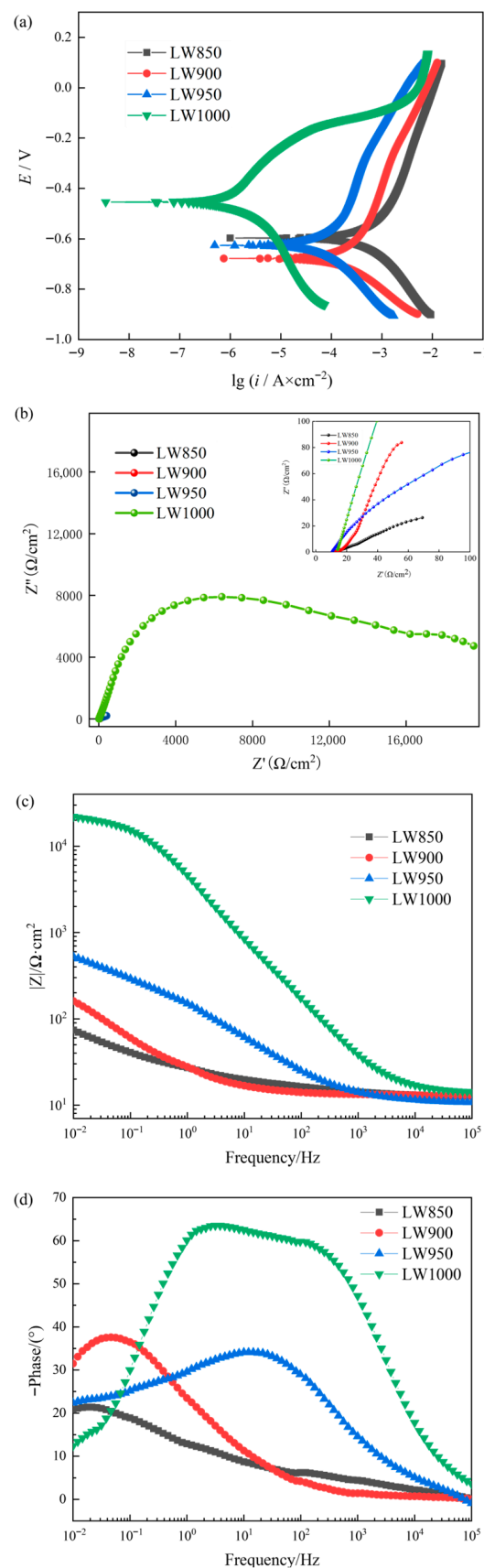


Figure 11. (a) Potentiodynamic polarization curves, (b) Nyquist plots, and (c,d) Bode plots in 3.5 wt% NaCl. LW1000 exhibits the best corrosion resistance, characterized by passivation, the largest capacitive arc, and the highest low-frequency impedance.

The excellent corrosion resistance of LW1000 is closely associated with its phase composition, microstructural evolution, and the stability of the surface oxide film, as revealed by combined XRD, SEM, and EDS analyses. Spinel structures, such as MgAl_2O_4 and FeAl_2O_4 , possess outstanding chemical stability and corrosion resistance. These phases contribute to the formation of a continuous and dense oxide film on the alloy surface, effectively blocking the penetration of Cl^- ions. While TiO is present in the alloys sintered at low-to-medium temperatures and acts as an anodic dissolution site due to its lower corrosion potential, it dissolves or transforms into more stable oxides during sintering at 1000°C , thereby reducing active sites for localized corrosion. The high-temperature sintering process promotes elemental diffusion and phase equilibration, minimizing compositional segregation and the inhomogeneous distribution of secondary phases. A more uniform microstructure implies fewer galvanic corrosion cells, consequently leading to a lower corrosion rate.

Figure 11b presents the Nyquist plots of the alloys. The corrosion resistance of the alloys is positively correlated with the diameter of the capacitive arc on the Nyquist curve. As shown in Figure 11b, the diameter of the capacitive arc increases with rising sintering temperature, indicating enhanced corrosion resistance. LW850, LW900, and LW950 exhibit a single capacitive arc, while LW1000 displays two capacitive arcs. Furthermore, the arc diameter for the high-temperature (1000°C) sintered alloy LW1000 is significantly larger than that of the alloys sintered at medium-low temperatures ($850\text{--}950^\circ\text{C}$). The single capacitive arc observed for LW850, LW900, and LW950 suggests that the corrosion process is primarily controlled by a single time constant, corresponding to charge transfer occurring on a heterogeneous surface. The phase composition of these alloys consists of a BCC matrix with various dispersed secondary phases (e.g., MgV_2O_4 , TiO). This structure does not allow the formation of a complete, dense, and protective surface film, leaving the alloy surface relatively “exposed.” Consequently, corrosion occurs directly at the metal/solution interface, which promotes pitting mechanisms [30]. High-temperature sintering promotes the surface enrichment of elements such as Al and Fe in LW1000. These elements combine with oxygen to form a dense, stable, and continuous (spinel-type) oxide film in situ on the alloy surface. The corrosion process thus proceeds in two steps: first, in the high-frequency region (the first capacitive arc), aggressive ions (e.g., Cl^-) undergo a slow transport process diffusing through the surface film. Second, in the low-frequency region (the second capacitive arc), only the few ions that manage to penetrate the film reach the underlying metal substrate, initiating substrate metal dissolution (charge transfer). The appearance of the double capacitive arc confirms the formation of a high-quality protective passive film on the LW1000 surface, shifting the corrosion mechanism from “direct attack” to a “penetration and defense” mode.

Figure 11c,d shows the Bode plots of the alloys. From the phase-angle plot in Figure 11c and the impedance modulus plot in Figure 11d, it can be observed that the phase angle and the impedance modulus in the low-frequency region for LW1000 are significantly greater than those of the other alloys. It has been shown that a higher impedance modulus in the low-frequency region indicates a lower corrosion rate and better corrosion resistance. The phase angle is typically related to capacitive behavior; a larger phase angle suggests less surface charge accumulation and higher capacitive performance of the alloy. These observations demonstrate that LW1000 exhibits the lowest corrosion rate and the best corrosion resistance, which is consistent with the analysis results obtained from the potentiodynamic polarization curves.

The EIS results of the MgAlTiVFeCo high-entropy alloy were fitted using equivalent circuits in Zview software to evaluate its corrosion resistance. The EIS data of LW850, LW900, and LW950 were fitted with the $R_s\text{--}(Q_{dl}/R_{ct})$ equivalent circuit model (Figure 12a),

while the EIS data of LW1000 were fitted with the $R_s-(Q_f//R_f)-(Q_{dl}//R_{ct})$ equivalent circuit model (Figure 12b). Here, R_s represents the solution resistance, R_f the passive film resistance, R_{ct} the charge transfer resistance, and Q_f and Q_{dl} the capacitance of the passive film and the double layer, respectively. A constant phase element (CPE) is commonly used to represent non-ideal capacitive behavior. The impedance of a CPE can be expressed as

$$Z_{CPE} = Y^{-1}(j\omega)^{-n} \quad (5)$$

where Y is the proportionality factor, n the dispersion coefficient, j the imaginary unit, and ω the angular frequency.

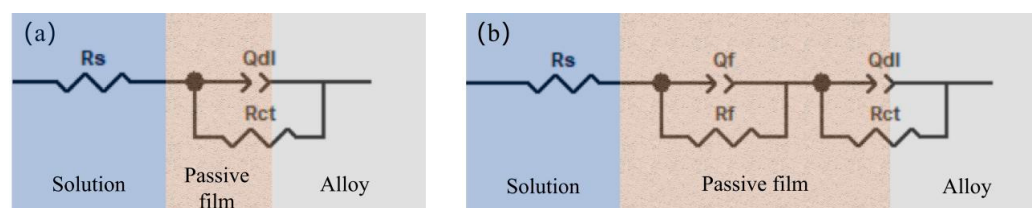


Figure 12. Equivalent circuits for fitting the EIS data: (a) $R_s-(Q_{dl}//R_{ct})$ model for LW850-LW950 and (b) $R_s-(Q_f//R_f)-(Q_{dl}//R_{ct})$ model for LW1000, indicating the presence of a protective passive film.

The fitting results are presented in Table 6. The corrosion process of the alloys prepared at low-to-medium temperatures (LW850, LW900, LW950) is governed by a single time constant, which corresponds to the charge transfer at the relatively exposed alloy/solution interface. The magnitude of the R_{ct} value directly reflects the difficulty of the electrochemical corrosion reaction. The fitting results show that as the sintering temperature increases from 850 °C to 950 °C, the R_{ct} value increases significantly from 159.80 $\Omega \cdot \text{cm}^2$ to 695.10 $\Omega \cdot \text{cm}^2$, indicating a gradual increase in the kinetic resistance of the corrosion reaction and an improvement in corrosion resistance. However, these values remain at the order of $10^2 \Omega \cdot \text{cm}^2$, which is relatively low and corresponds to the higher corrosion current densities observed in the polarization curves. The Q_{dl} value of the alloys drops sharply from $4.78 \times 10^{-2} \Omega^{-1} \cdot \text{cm}^{-2} \cdot \text{s}^n$ for LW850 to $2.66 \times 10^{-3} \Omega^{-1} \cdot \text{cm}^{-2} \cdot \text{s}^n$ for LW950. Q_{dl} is related to the true electrode surface area and the dielectric properties of the interfacial layer; its decrease by an order of magnitude indicates a reduction in the active area available for the corrosion reaction. Meanwhile, the n_1 values range between 0.46 and 0.52, significantly less than 1, indicating that the double layer exhibits strong non-ideal capacitive behavior (dispersion effect). This arises from the high heterogeneity of the alloy surface—a multiphase structure composed of the BCC matrix phase and various dispersed secondary phases (e.g., MgV_2O_4 , TiO)—resulting in non-uniform current and potential distribution and the absence of a complete, dense protective surface film. Corrosive ions (e.g., Cl^-) can readily reach the metal surface, triggering a “direct-attack” corrosion mechanism characterized by localized pitting.

Table 6. Equivalent Circuit Fitting Parameters.

Alloy	R_s ($\Omega \cdot \text{cm}^2$)	Q_{dl}		R_{ct} ($\Omega \cdot \text{cm}^2$)	Q_f		R_f ($\Omega \cdot \text{cm}^2$)
		Y_{dl} ($\Omega^{-1} \cdot \text{cm}^{-2} \cdot \text{s}^n$)	n_1		Y_f ($\Omega^{-1} \cdot \text{cm}^{-2} \cdot \text{s}^n$)	n_2	
LW850	20.02	4.78×10^{-2}	0.46	159.80	--	--	--
LW900	13.02	2.37×10^{-2}	0.52	688.80	--	--	--
LW950	9.69	2.66×10^{-3}	0.47	695.10	--	--	--
LW1000	13.92	2.28×10^{-4}	0.71	85.42	5.00×10^{-5}	0.78	2.29×10^4

The EIS of the high-temperature alloy LW1000 exhibits distinct dual capacitive arc characteristics. The fitting yields an extremely high passive film resistance, $R_f = 2.29 \times 10^4 \Omega \cdot \text{cm}^2$, which is 1–2 orders of magnitude greater than the R_{ct} values of the medium- and low-temperature alloys. This demonstrates that after sintering at 1000 °C, a dense, continuous, and highly stable protective oxide film (an Al- and Fe-enriched (spinel-type) structure) has formed in situ on the alloy surface. This film acts as a robust physical and electrochemical barrier, significantly hindering the penetration and transport of aggressive ions (Cl^-). Concurrently, the passive film capacitance, Q_f ($5.00 \times 10^{-5} \Omega^{-1} \cdot \text{cm}^{-2} \cdot \text{s}^n$), is much smaller than the Q_{dl} values of the medium- and low-temperature alloys, and its dispersion coefficient, $n_2 = 0.78$, is closer to 1. This indicates that the film possesses relatively uniform and dense dielectric properties. Under the protection of this passive film, the amount of corrosive medium reaching the substrate interface is significantly reduced, leading to a further decrease in the double-layer capacitance at this interface to $Q_{dl} = 2.28 \times 10^{-4} \Omega^{-1} \cdot \text{cm}^{-2} \cdot \text{s}^n$. However, the corresponding charge transfer resistance, R_{ct} ($85.42 \Omega \cdot \text{cm}^2$), is far lower than that of LW950. This suggests that once the corrosive medium penetrates the dense passive film through micropores or defects and contacts the fresh metal substrate, localized corrosion reactions can still proceed readily. Nevertheless, the extremely low Q_{dl} and very high R_f collectively indicate that the number and area of such penetration sites are minimal, and the path for ions through the film is greatly obstructed. Therefore, the overall corrosion rate is dominated by ion diffusion within the film (controlled by R_f), rather than by the charge transfer reaction beneath the film.

Figure 13 shows the surface corrosion morphology of the MgAlTiVFeCo high-entropy alloy after potentiodynamic polarization testing in a 3.5 wt% NaCl solution. As shown in Figure 13, the corrosion morphology of the LW850 and LW900 alloys exhibits typical severe general corrosion characteristics. The surface is extensively covered by a loose, non-uniform layer of corrosion products, with densely distributed deep and large pits visible. The corrosion morphology of the LW950 alloy shows some improvement compared to LW850 and LW900, but overall, it still belongs to non-uniform corrosion. The surface displays large, lumpy accumulations of corrosion products, while in some areas the substrate metal is exposed, revealing features of localized dissolution. It is evident that the increase in sintering temperature refines the microstructure to some extent or promotes surface diffusion of certain elements, slightly mitigating the surface chemical heterogeneity and reducing the severity of corrosion. The surface of the LW1000 alloy is flat and smooth, with only a few small, isolated pitting spots scattered across it, demonstrating excellent localized corrosion characteristics. High-temperature (1000 °C) sintering promotes the surface enrichment of elements such as Al and Fe, leading to the formation of a dense, continuous (spinel-type) passive film. This dense passive film (high R_f) effectively blocks the uniform penetration of aggressive Cl^- ions, confining corrosion to local weak points in the film. Once the film is locally ruptured and aggressive ions reach the substrate, pitting is initiated, forming the isolated pits seen in the figure. However, due to the extremely strong protective nature of the film (high R_f), the pits are prevented from propagating and connecting, thereby preserving the majority of the substrate. The extremely low double-layer capacitance Q_{dl} also indirectly confirms that the true area undergoing corrosion reactions is very small.

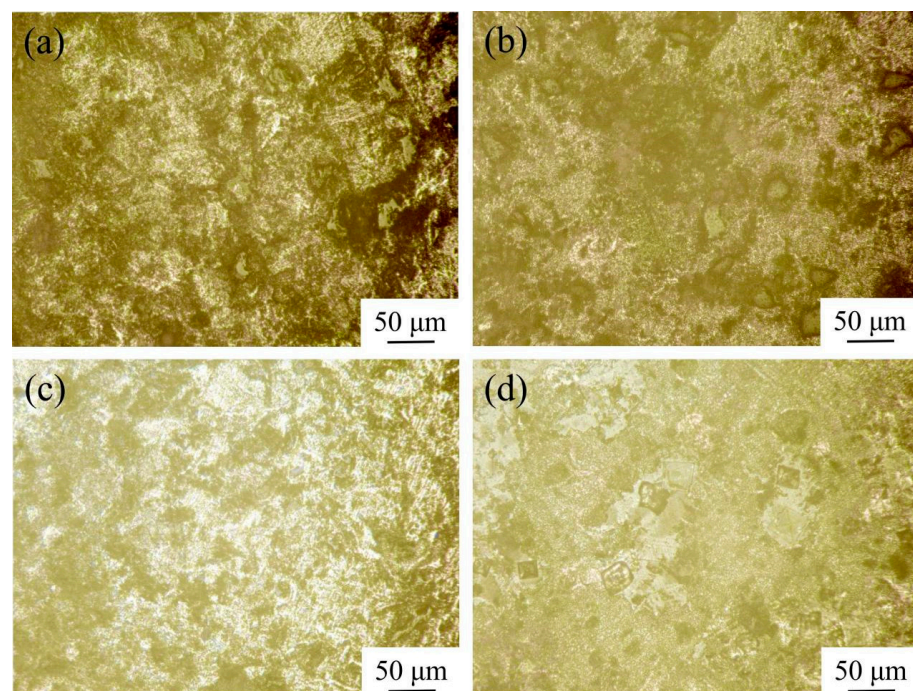


Figure 13. SEM images showing corrosion morphology after polarization tests. Corrosion damage transitions from severe general attack (LW850/LW900) to localized pitting on a largely intact surface (LW1000): (a) LW850, (b) LW900, (c) LW950, (d) LW1000.

4. Potential Applications

Owing to its unique combination of properties—low density (theoretical density: 5.304 g/cm³), high hardness (up to 1235.5 HV), excellent high-temperature oxidation resistance, and good corrosion resistance in chloride-containing environments—the MgAl-TiVFeCo lightweight high-entropy alloy prepared in this study demonstrates promising potential for applications in several high-technology fields. In the aerospace sector, the alloy could serve as weight-sensitive fasteners, brackets, or peripheral engine components, achieving a balance between lightweight design and high specific strength. In the lightweight design of transportation vehicles, its high specific hardness and wear resistance are valuable attributes. For marine engineering and chemical equipment, its outstanding resistance to chloride-ion corrosion makes it suitable for pumps, valves, ship components, and certain reactor linings. Furthermore, it holds potential for lightweight tools or molds requiring certain thermal stability, such as dies for aluminum alloy die-casting. This work aims to clarify the effect of sintering temperature on the microstructure and properties of the alloy, thereby providing critical processing and performance data to support its transition toward practical engineering applications.

5. Conclusions

In this work, a six-component MgAlTiVFeCo lightweight high-entropy alloy was successfully prepared by mechanical alloying (MA) combined with spark plasma sintering (SPS). The alloying behavior of the powder during ball milling, as well as the influence of sintering temperature (850–1000 °C) on the microstructure, densification behavior, mechanical properties, high-temperature oxidation resistance, and corrosion resistance of the alloy were systematically investigated. The main conclusions are as follows:

- (1) Nanocrystalline BCC solid-solution precursor powders were synthesized via mechanical alloying. During SPS, the BCC matrix was retained while the precipitation of various protective oxide second phases (predominantly spinel-type) was promoted.

Sintering temperature serves as a key parameter governing the composition, content, and spatial distribution of the second phases. At 1000 °C, a microstructure most conducive to property optimization is achieved.

- (2) Both the relative density and microhardness of the alloy exhibit a synergistic increasing trend with rising sintering temperature. Sintering at 1000 °C achieves a combination of high relative density (96.7%) and high hardness (1235.5 HV). This strengthening results from the coupling of multiple mechanisms, including precipitation strengthening by oxide second phases and grain-boundary pinning, leading to grain refinement.
- (3) The oxidation kinetics of the alloy in air at 900 °C follow a parabolic law and are diffusion-controlled. Sintering temperature significantly influences oxidation resistance by altering the relative density, distribution of second phases, and grain-boundary structure. The sample sintered at 1000 °C (LW1000) exhibits the most outstanding oxidation resistance due to its dense microstructure and uniformly distributed barrier phases that effectively impede diffusion.
- (4) In a 3.5 wt% NaCl solution, the corrosion resistance of the alloy improves with increasing sintering temperature. The sample sintered at 1000 °C (LW1000) exhibits the most positive corrosion potential (−0.438 V) and the lowest corrosion current density ($1.07 \times 10^{-6} \text{ A} \cdot \text{cm}^{-2}$). High-temperature sintering promotes the formation of a dense, stable, and high-resistance passive film (enriched with Al- and Fe-bearing spinel structures), shifting the corrosion mechanism from “active dissolution” to “medium permeation-passive film barrier”. This transformation significantly impedes the attack of Cl^- ions, thereby achieving excellent corrosion resistance.

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